

JAYADEEP GOWDA K B

Robotics - AI Software Systems Engineer

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EDUCATION

Bangalore Institute of Technology

B.E. Robotics & Artificial Intelligence | CGPA: 8.88/10

Bangalore, India

2023 – 2027

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL
- **Software Systems & Methodologies:** Git, APIs, OpenCV, YOLO, Linux/ARM Dev
- **ML/DL & AI:** PyTorch, TensorFlow, OpenVINO, ONNX, Transformers, CNN-LSTM Systems, Multimodal Pipelines
- **Robotics Engineering:** ROS2, Kinematics, Webots, Gazebo, ABB 6-DOF ARM, ABB Robot Studio

EXPERIENCE

Indian Institute of Science (IISc)

ML Research Intern – Dept. of Computational & Data Sciences

Bangalore, India

Mar 2026 – Jun 2026

- Developed end-to-end analytical data pipelines to ingest and process high-frequency material sensor data from advanced MXene piezoresistive wristbands, handling disparate sampling rates (5.68–26.32 Hz).
- Benchmarked 17 diagnostic models (14 classical ML + CNN, BiLSTM, CNN-LSTM hybrid architectures) using 5-fold stratified cross-validation and 21 mathematical time/frequency domain features.
- Collaborated directly with Postdocs, PhD candidates, and International research partners (AGH University, Poland) on advanced scientific studies.

Sirena AI Systems

R&D Intern – Automation & Edge AI

Bangalore, India

Jan 2026 – Feb 2026

- Spearheaded the full-lifecycle engineering of a production-grade desktop robot, executing system requirements from virtual physics simulations to multi-DOF physical embedded hardware deployment.
- Applied and evaluated complex locomotive systems across Humanoid, Quadruped, and Hexapod robotic structures using PyBullet, developing custom inverse kinematics solvers and 12-state FSM gait controllers.
- Integrated an optimized, real-time perception pipeline combining YOLOv8, Whisper speech processing, and on-device Phi-2 LLM execution, reducing inter-process communication latency to under 50ms via ZeroMQ pub-sub on ARM targets.

Frillory Design House

AI Automation Engineer

Bangalore, India

Jun 2025 – Aug 2025

- Architected an agentic software automation toolchain leveraging LLM prompt orchestration and generative asset pipelines; automated 50+ distinct enterprise assets, reducing manual overhead by 25%.

PROJECTS

MEMO (Neural-OS) – Multimodal Autonomous Desktop Robot Agent

GitHub

- Designed and manufactured an interactive desktop robot assistant driven by an onboard Raspberry Pi 5 executing synchronized compute pipelines for vision (YOLOv8, FaceNet), speech (Whisper), and localized language logic.
- Optimized low-level concurrent text-to-speech execution and compressed 4-bit quantized 2.7B parameter models to perform at 10–14 tokens/sec, reducing end-to-end latency to 600ms via targeted script optimization.
- Presented technical implementation details and real-time interaction capabilities at national expo.

Awarded 3rd Place for Best Poster Presentation.

YOLOmart – Smart Retail Infrastructure & Tooling

GitHub

- Built and deployed a compressed data-driven software backend using extreme optimization workflows (INT8 Quantization), minimizing model payload requirements from 100MB down to 3.5MB.
- Engineered a robust analytical camera-to-dashboard pipeline operating reliably at 20 FPS with 92% mAP under dynamic environments; built interactive data flows for mobile dashboards.

Awarded 2nd Place, GlitchVerse 2k25.

LEADERSHIP & RECOGNITION

- **Competitions:** 2nd Place out of nationwide teams at GlitchVerse 2k25 Project Expo | 3rd Place for Best Poster Presentation (On Memo) | National Finalist at Anveshana 2025 Engineering Competition.
- **Leadership:** Joint Secretary of ROBO CELL Club; coordinated heads, managed cross-functional workflows served as Core Organizer for Ideathon 2025 hosting 600+ participants.